

Eric J. Leonardis, MS, PhD

San Diego, CA. 92110

Email: leonardiseric@gmail.com

Phone: 516-510-2096

SKILLS

Programming: Python, Matlab, Arduino, Processing, Bash/Shell Scripting

Packages: JAX, Pytorch, TensorFlow, Mujoco, DeepMind Control, DeepMind Acme, Ray

Machine Learning: Reinforcement Learning, Imitation Learning, Deep Learning, Pose Estimation and Tracking

Signal Processing: Time–frequency analysis, filters, nonlinear dynamical systems, state-space reconstruction methods

Hardware: Neural implants, cameras, sensors, actuators, embedded systems

Software Development: Git, CI/CD, pytest, agentic workflows, prompt engineering

EDUCATION AND TRAINING

Salk Institute for Biological Studies – August 2022 to August 2026

Postdoctoral Fellow in Computational Neurobiology – NIH T32 Trainee, Foresight Institute Trainee

Research Interests: Deep Reinforcement Learning, 3D Pose Estimation, Biomechanical Physics Simulations

University of California, San Diego (UCSD) – September 2014 to June 2022

Ph.D. in Cognitive Science – Systems Neuroscience, Robotics

M.S. in Cognitive Science – Brain-Computer Interfaces

Hofstra University – Fall 2010 to Spring 2014

B.A. in Psychology

EXPERIENCE

Principal Scientist, Autonomous Cardiac Control Systems (MaxWell Biomedical) – Aug 2026 – Present

Developing a next generation autonomous control system for cardiac resynchronization therapy using techniques from dynamical systems theory, signal processing, electrophysiology, and machine learning.

Machine Learning and 3D Computer Vision Consultant (UCSD) – Jan 2026 – Aug 2026

Training and supporting technical teams in deploying open-source computer vision tools to transform multi-camera video data into structured behavioral signals for analyzing engagement and interaction patterns in educational environments.

Postdoctoral Fellow – Talmo Pereira Lab and Eiman Azim Lab (Salk) – Aug 2022 – Aug 2026

Developed JAX-based pipelines for 3D pose estimation, musculoskeletal simulation, and imitation learning to study mouse motor control. Built and benchmarked physics-aware control objectives linking simulated muscle activations to in vivo EMG.

Graduate Researcher – Andrea Chiba Lab (UCSD) – August 2015 – Aug 2022

Developed reinforcement learning–based robotic paradigms for studying rodent–agent interaction. Led international multidisciplinary engineering teams and large-scale behavioral and neural data collection and analysis.

Visiting Scholar – Complex and Intelligent Systems Lab (UQ) – Brisbane, AU – March 2016, November 2017

Worked with Professor Janet Wiles and her team at the University of Queensland to create novel methods for dynamical systems analysis and topological data analysis.

PROJECTS

Musculoskeletal Imitation Learning of Mouse Forelimb Reaching (MIMIC-MJX)

First author on a NeurIPS workshop paper introducing a high-throughput imitation learning framework for dexterous mouse forelimb reaching, showing that physics-informed regularization yields simulated muscle activations better aligned with in vivo EMG. Co-author on the broader platform paper, MIMIC-MJX: Neuromechanical Emulation of Animal Behavior (under review at Nature Methods), an open-source pipeline integrating motion capture (STAC-MJX) with massively parallel imitation learning (Track-MJX) in a musculoskeletal MuJoCo-MJX model, validated against experimental kinematics using GPU-accelerated JAX rollouts. ArXiv: <https://arxiv.org/pdf/2511.21848> | <https://arxiv.org/pdf/2511.20532>

LUC3D: Browser-Based 3D Annotation and Proofreading GUI for Pose Estimation and Multi-Animal Tracking

First author of LUC3D (In Prep), a zero-install browser-based GUI that reduces the manual labor of labeling, proofreading, and multi-camera multi-animal 3D tracking. Released alongside comprehensive proofread 3D triangulated pose datasets of multi-animal mouse social behavior, a protocol for generating 3D-consistent labels, and a cross-view ReID tracking algorithm, lowering the cost of entry for 3D pose estimation in neuroscience labs. Code: <https://github.com/talmolab/luc3d> | Docs: <https://talmolab.github.io/luc3d/>